E-News Express

Business Statistics

06/09/2025

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Executive Summary

Conclusions

- The data analysis shows that the new landing page of E-News Express has increased the mean spending time than the old page.
- Also, proportionally, more users of the new landing page get converted to subscription than the old landing page and the users who get converted also spend more time on the landing page.
- Preferred language does not impact the conversion to subscription.
- Mean time spent on the landing pages is not dependent on the preferred language. In other words, all English, French and Spanish users spend same amount of time on the landing page.

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Executive Summary

Actionable insights

- The new landing page seems to be working well and much preferred. Users are spending more time on it and getting more subscriptions. There may be need for a survey for understanding which features are working well and are there any other items to improve.
- Are there any other variables that the conversion rate depends on? It may be good to get survey data about age, sex and education of users and see if any of them are correlated with the conversion to subscription.
- Preferred language does not seem to be making any difference as far as the conversion rate or time spent on the landing pages.

Recommendations

- Enable all users to use the new landing page, as it's design seems to be better than the old page.
- Enhance subscription models to encourage long-term engagement.
- Conduct surveys to obtain some more variables for data analysis and any suggestions from users..

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Business Problem Overview and Solution Approach

- Problem
  - E-news Express, an online news portal, aims to expand its business by acquiring new subscribers
  - The information on these portals

    * is retrieved electronically from online databases

    * processed using a variety of software, and

    * transmitted to the users
- Solution - Design team has a new landing webpage.
  - Data provided is a random sample of 100 users using the old and new landing pages.
  - This data needs to be analyzed to get insights about effectiveness of the new landing page, time spent by users on landing pages and conversion rate
  - The results of the analysis can give direction to business expansion

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Business Problem Data Analysis Goals

To determine the effectiveness of the new landing page in gathering new subscribers for the news portal by answering the following questions:

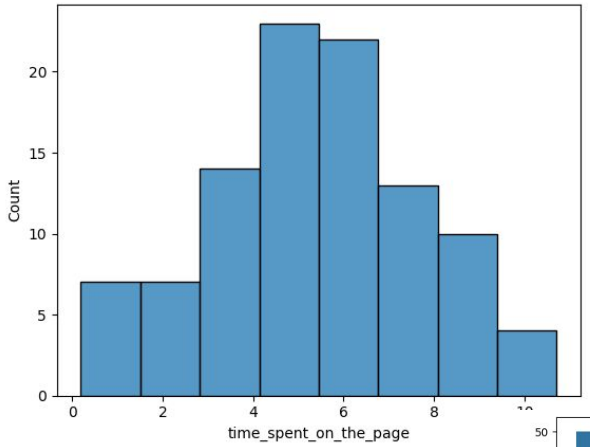
1. Do the users spend more time on the new landing page than on the existing landing page?
2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?
3. Does the converted status depend on the preferred language? [Hint: Create a contingency table using the pandas.crosstab() function]
4. Is the time spent on the new page the same for the different language users?

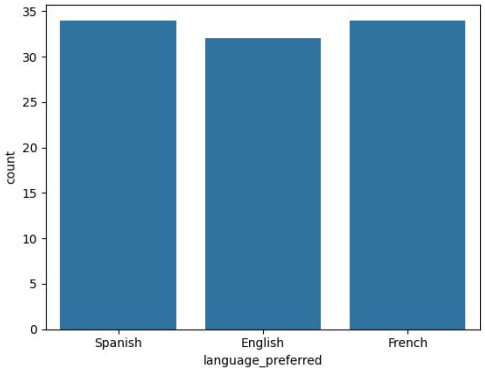
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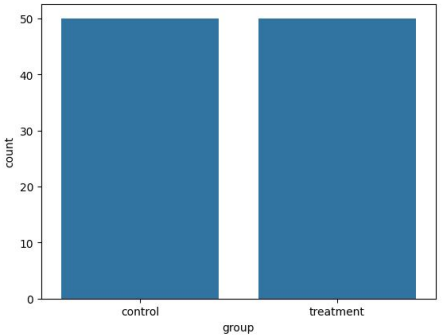
EDA Results- Data Overview

The data contains information regarding the interaction of users in both groups with the two versions of the landing page.

1. user_id - Unique user ID of the person visiting the website
2. group - Whether the user belongs to the first group (control) or the second group (treatment)
3. landing_page - Whether the landing page is new or old
4. time_spent_on_the_page - Time (in minutes) spent by the user on the landing page
5. converted - Whether the user gets converted to a subscriber of the news portal or not
6. language_preferred - Language chosen by the user to view the landing page

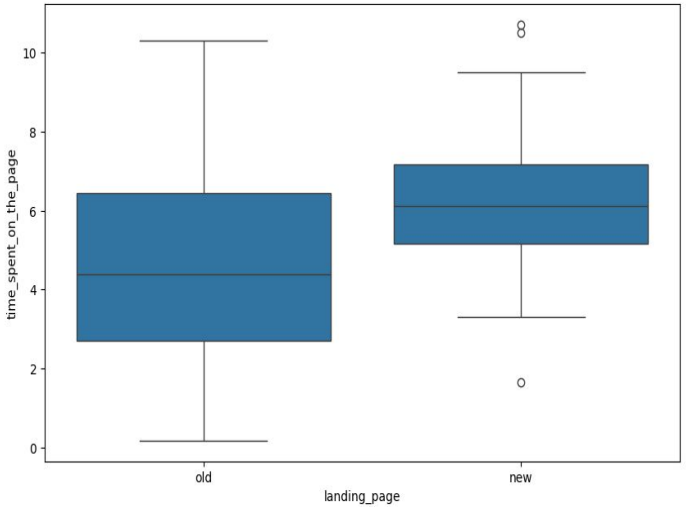


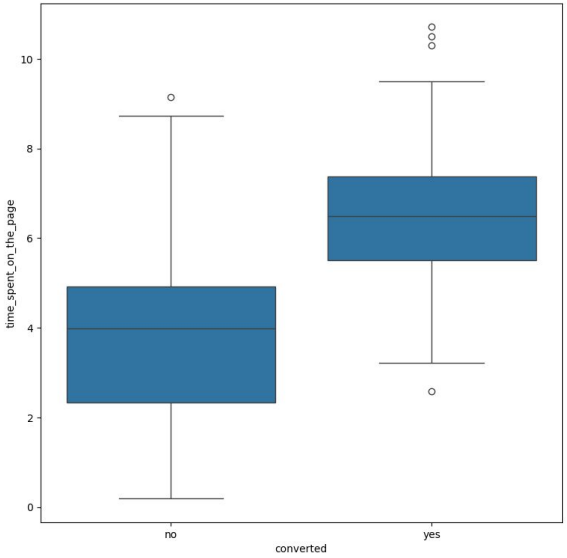


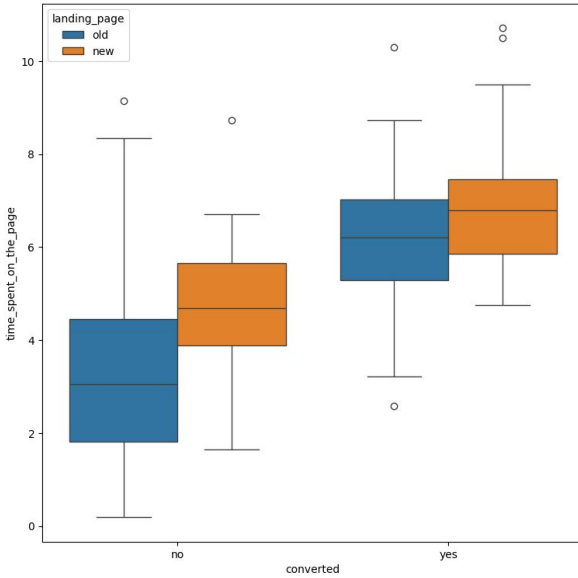


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EDA Results- Time spent on old and new pages and conversion



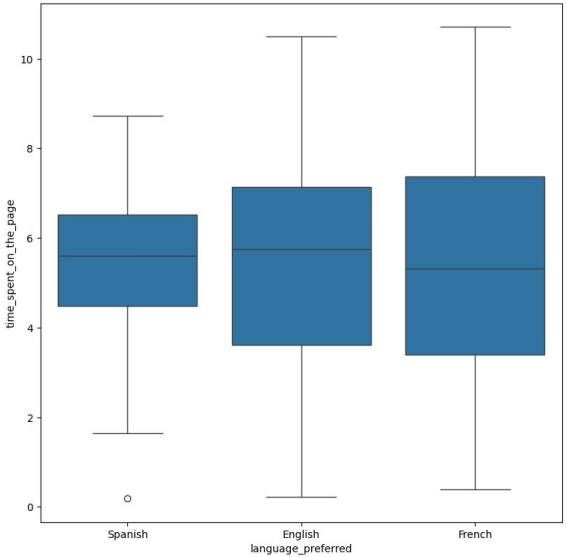


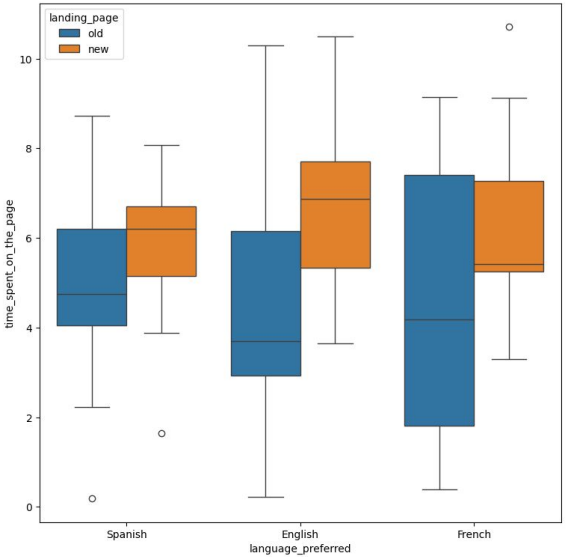


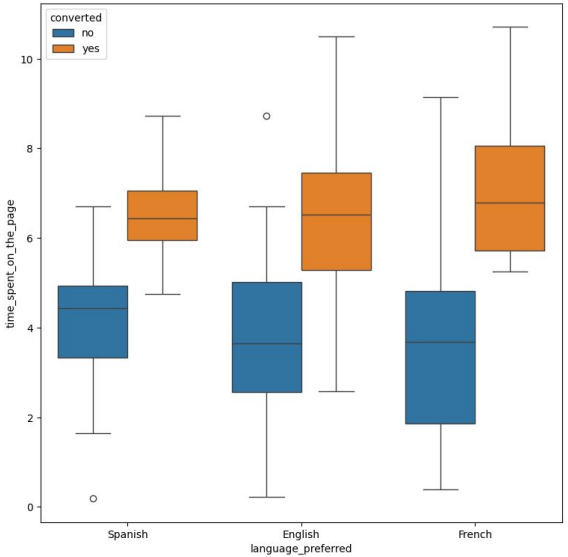
- *The users seem to spend more time on the new landing page rather than the old landing page.*
- *The time spent by the converted users is more than non-converted users regardless of the page*
- *The users who spend more time on the landing page (old or new), seem to get converted to subscription*

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EDA Results- Language preferences and time spent on pages







- *The average time spent by English users on the landing pages is slightly more than Spanish or French users. But they are all almost equal*
- *The new landing page users from all languages spend more time on the page*
- *The converted users spend much more time on the page regardless of preferred language.*

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Statistical analysis

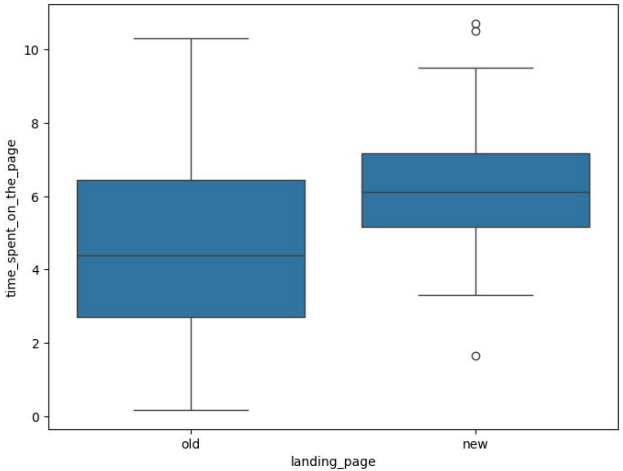
For the next four statistical tests, level of significance was determined to be 0.05

Also, for checking for normalization, Levene test was used for confirming equality of variances.

Population variances were found to be equal.

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Time spent on New Vs Old landing page



Hypothesis

H_0: The mean time spent on the old landing page is equal to the mean time spent on the new landing page.

H_a: The mean time spent on the new landing page is greater than the mean time spent on old landing page.

Test result and inference

This is a one-tailed test concerning two population means from two independent populations. The population standard deviations are unknown. **Two sample independent t-test can be used.**

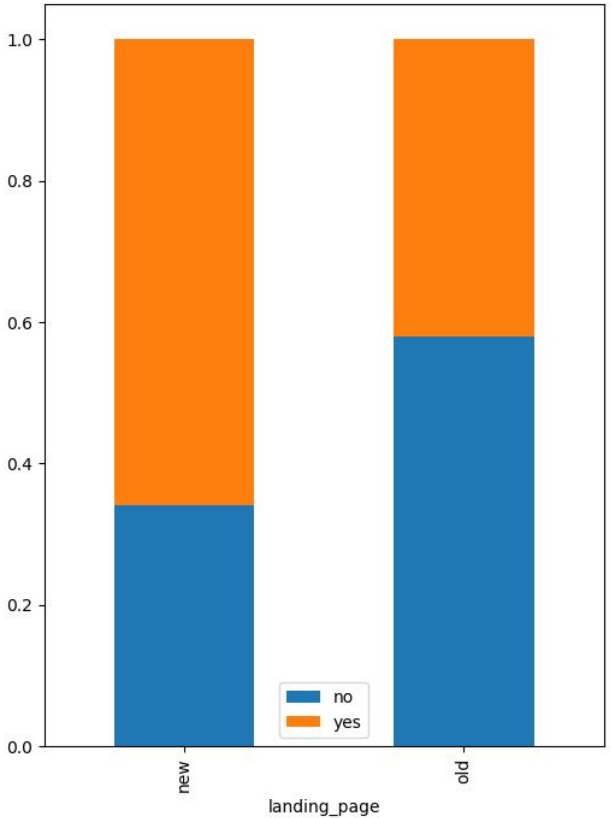
Data is divided into two subsets - for old and new landing page.

```
As the p-value 0.0001316123528095005 is less than the level of
significance 0.05, we reject the null hypothesis.
```

There is enough statistical evidence to say that the mean time spent on the new landing page is more than the mean time spent on the old landing page.

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Conversion rate of New Vs Old landing page users



Hypothesis

H_0: Conversion rate proportion for the old page and new page is same.

H_a:Conversion rate proportion of the new page is more than the old page.

Test result and inference

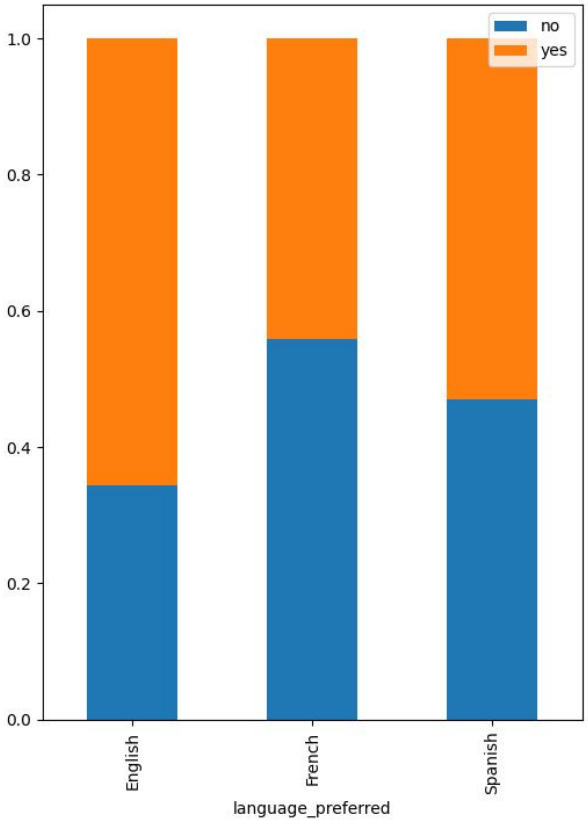
This is a one-tailed test concerning two population proportions from two independent populations. **Two sample proportions Z test is used.**

```
As the p-value 0.008026308204056278 is less than the level of
significance 0.05, we reject the null hypothesis.
```

There is enough statistical evidence to say that the conversion rate proportion of the new landing page is more than the conversion rate proportion of the old landing page.

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Does converted status depend on the language preference?



Hypothesis

H_0: Converted status is independent of preferred language.

H_a: Converted status is dependent on preferred language.

Test result and inference

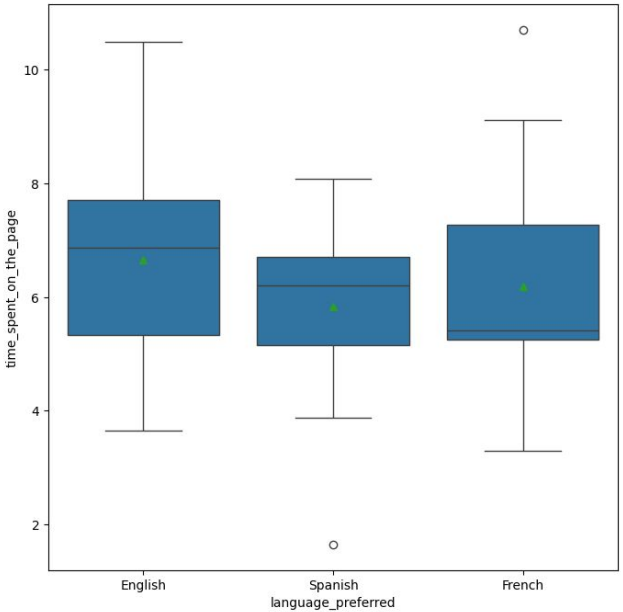
This is a problem of the test of independence, concerning two categorical variables - converted status and preferred language. **Chi square test of independence is chosen**. Contingency table of Converted Vs Language preferred columns is created.

```
As the p-value 0.2129888748754345 is greater than the level of
significance 0.05, we fail to reject the null hypothesis.
```

Our data does not show that Converted status is dependent on language preferred as it is not statistically significant.

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Is the time spent on new page same for diff language users?



Hypothesis

H_0: Average time spent on the new page is same for all languages users

H_a: Average time spent on the new page is different for at least one language users group from the other languages

Test result and inference

This is a problem, concerning three population means.**One way ANOVA test was used. Levene test can be used for confirming equality of variances.** A subset of data only for new landing page needs to be used.

```
As the p-value 0.43204138694325955 is greater than the level of
significance 0.05, we fail to reject the null hypothesis.
```

There is not enough statistical significance to conclude that mean time spent on page is different for at least one preferred language out of English, Spanish and French.

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